Programming Homework: Bayesian Learning with Scikit-learn

Problem 1: Comparing MLE vs MAP in Linear Regression

Requirements:

Compare Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP) estimation for linear regression:

a) Create synthetic data:

- Randomly generate 100 data points: $X \sim \mathcal{N}(0, 1)$
- Generate targets: $y = 2X + 1 + \epsilon$, where $\epsilon \sim \mathcal{N}(0, 0.5)$
- Split into train/test sets (70/30 ratio)

b) Perform MLE estimation (Linear Regression):

- Use `LinearRegression` from scikit-learn
- Calculate the regression coefficients and intercept

c) Perform MAP estimation (Ridge Regression):

- Use `Ridge` from scikit-learn
- Experiment with different α values: $[0.1, 1.0, 10.0, 100.0]$
- Explain the relationship: $\alpha = \frac{\sigma^2}{\tau^2}$

d) Evaluation and comparison:

- Calculate MSE on the test set for both methods
- Compare the estimated coefficient values
- Plot the regression lines and data points
- Analyze the effect of α on the magnitude of the coefficients

e) Experiment with noisy data:

- Add outliers to the training set (5% of data points with large noise)
- Compare the robustness of MLE and MAP

Problem 2: Bayesian Classification and Naive Bayes

Requirements:

Perform text classification using Bayesian methods:

a) Data preparation:

- Use the `20newsgroups` dataset from scikit-learn
- Select 2 categories: `sci.med` and `sci.space`
- Split into train/test sets (80/20 ratio)

b) Text preprocessing:

- Use `CountVectorizer` or `TfidfVectorizer`
- Limit the maximum number of features (max_features=5000)
- Apply TF-IDF transformation

c) Train Bayesian models:

- `GaussianNB`: Naive Bayes with Gaussian likelihood
- `MultinomialNB`: Naive Bayes with Multinomial likelihood
- `BernoulliNB`: Naive Bayes with Bernoulli likelihood
- `ComplementNB`: A variant of MultinomialNB for imbalanced data

d) Performance evaluation:

- Calculate Accuracy, Precision, Recall, and F1-score on the test set
- Plot the confusion matrix for each model
- Compare training and inference times

e) Analyze the effect of Laplace Smoothing:

- Experiment with different α values in `MultinomialNB`
- Analyze the impact on performance and stability

f) Visualize important features:

- Find the words with the highest $P(\text{word}|\text{class})$ probability for each class
- Explain the results based on Bayes' theorem

References

- Scikit-learn Documentation: https://scikit-learn.org/stable/modules/naive_bayes.html
- Bayesian Learning Slides (BayesianLearning.pdf)
- "Pattern Recognition and Machine Learning" - Christopher M. Bishop
- "Bayesian Reasoning and Machine Learning" - David Barber